

# DAY 4 — GPUs AND COOLING SYSTEMS

## THEORY — 30 MINUTES

### 1. What is a GPU?

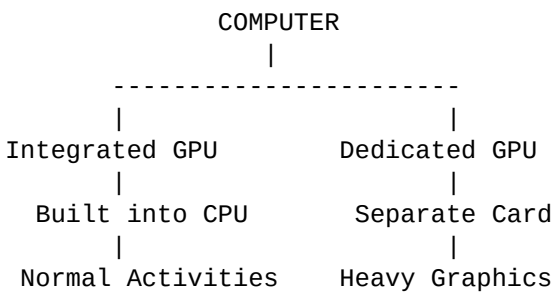
**GPU = Graphics Processing Unit**

A GPU is a computer component that processes images, videos, animations, games, and other graphics.

### Types of GPU

| Integrated GPU                         | Dedicated GPU                                         |
|----------------------------------------|-------------------------------------------------------|
| Built into the CPU/processor.          | A separate graphics card.                             |
| Uses the computer's RAM.               | Has its own VRAM.                                     |
| Good for normal work and light gaming. | Better for gaming, video editing, and heavy graphics. |

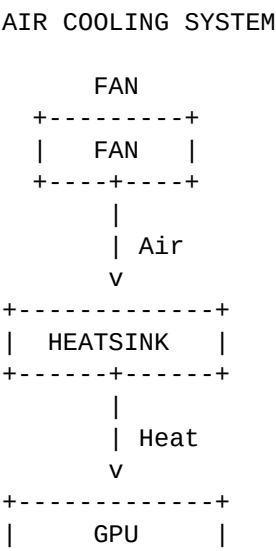
### GPU Diagram



## 2. GPU COOLING SYSTEM

A GPU produces heat when it is working. A cooling system removes this heat and helps keep the GPU at a safe temperature.

### AIR COOLING



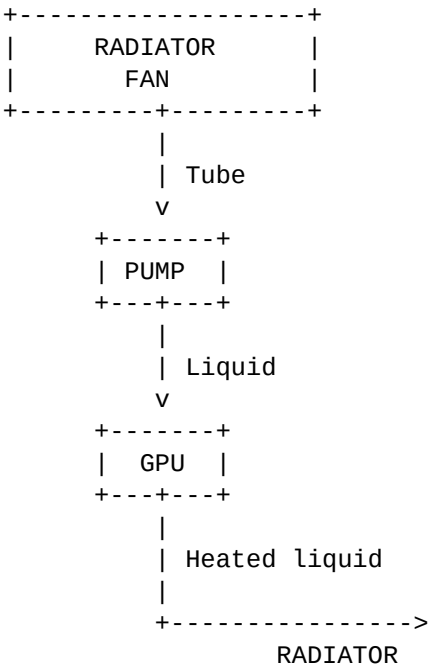
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Main Parts

| Part          | Function                                  |
|---------------|-------------------------------------------|
| Fan           | Moves air.                                |
| Heatsink      | Takes heat away from the GPU.             |
| Thermal paste | Helps transfer heat from GPU to heatsink. |
| GPU           | Processes graphics.                       |

LIQUID COOLING

LIQUID COOLING SYSTEM



| Part     | Function                 |
|----------|--------------------------|
| Pump     | Moves the liquid.        |
| Liquid   | Carries heat away.       |
| Tubes    | Carry the liquid.        |
| Radiator | Releases heat.           |
| Fan      | Helps cool the radiator. |

# PRACTICAL — 30 MINUTES

## TASK 1 — IDENTIFY YOUR GPU

Open your laptop or desktop and find the GPU information.

GPU Brand: \_\_\_\_\_

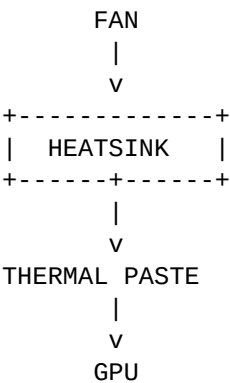
GPU Model: \_\_\_\_\_

Type: Integrated / Dedicated

Example:  
Brand: NVIDIA  
Model: GeForce RTX 4060  
Type: Dedicated

## TASK 2 — IDENTIFY THE COOLING SYSTEM

Take a clear photo or screenshot of the cooling system and label the main parts.



### Functions

**Fan:** Moves air and removes heat.

**Heatsink:** Absorbs and transfers heat away from the GPU.

**Thermal paste:** Helps transfer heat between the GPU and heatsink.

**GPU:** Processes graphics.

## TASK 3 — AIR COOLING VS LIQUID COOLING

| Feature      | Air Cooling       | Liquid Cooling                   |
|--------------|-------------------|----------------------------------|
| Cost         | Usually cheaper   | Usually more expensive           |
| Cooling      | Good              | Strong cooling                   |
| Maintenance  | Easy              | More complex                     |
| Installation | Easy              | More difficult                   |
| Main Parts   | Fan + heatsink    | Pump + liquid + tubes + radiator |
| Common Use   | Normal gaming PCs | High-performance PCs             |

## TASK 4 — GAMING + IPL STREAMING PC

My Choice: AIR COOLING

I would choose air cooling for my gaming and IPL streaming PC. It is generally cheaper and easier to install and maintain. A good GPU air-cooling system can provide sufficient cooling for gaming and streaming. It also has fewer components than a liquid-cooling system.

## QUICK REVISION

```
GPU
|
+-- Integrated GPU
|
+-- Dedicated GPU
```

```
COOLING
|
+-- Air Cooling
|   |
|   +-- Fan
|   +-- Heatsink
|
+-- Liquid Cooling
    |
    +-- Pump
    +-- Liquid
    +-- Tubes
    +-- Radiator
    +-- Fan
```

### Remember

|                       |                                             |
|-----------------------|---------------------------------------------|
| <b>GPU</b>            | Processes graphics                          |
| <b>Fan</b>            | Moves air                                   |
| <b>Heatsink</b>       | Removes heat                                |
| <b>Thermal paste</b>  | Transfers heat                              |
| <b>Pump</b>           | Moves liquid                                |
| <b>Radiator</b>       | Releases heat                               |
| <b>Air cooling</b>    | Simple and cheaper                          |
| <b>Liquid cooling</b> | More complex and can provide strong cooling |